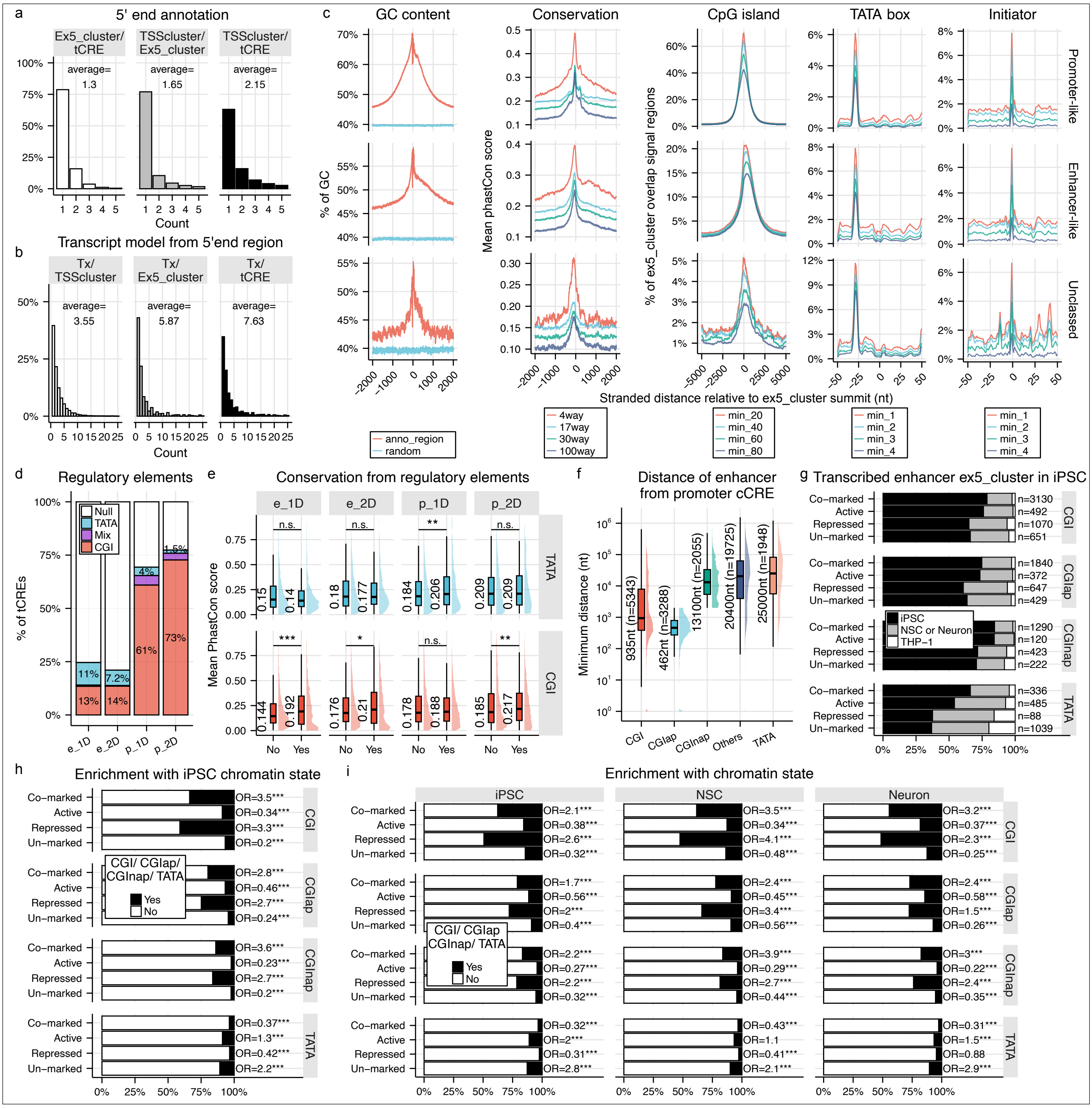

Supplementary Figure 1 | Features of extended 5′ end clusters (ex5_clusters)

a, Comparison between TSS clusters, ex5_clusters, and tCREs, showing average coverage of each feature across the others. **b**, Number of transcript models originating from each 5′ end feature (TSS cluster, ex5_cluster, and tCRE). **c**, Distribution of DNA regulatory elements associated with ex5_clusters (linked to transcript models in the finalized transcriptome, $n = 75{,}658$) in reference to summit position, categorized as promoter-like, enhancer-like, or unclassed. Both major and minor strand clusters were included. See Methods and **Extended Data Figure 2e** legend for classification criteria. **d**, Presence of CGI, TATA box, or both (mix) in 2D and 1D ex5_clusters, assessed using non-redundant 1001 nt windows. **e**, Conservation scores (phastCons 17-way) of ex5_clusters grouped as 2D or 1D promoters or enhancers, and further stratified by presence or absence of CGI or TATA box. Mean phastCons values were calculated within ±500 nt of cluster summits. **f**, Distance of different classes of enhancer from promoter cCRE. **g**, Coordinations of enhancer-like ex5_clusters (major strand only) were intersected with iPSC H3K27ac and H3K27me3 CUT&Tag data and grouped into Co-marked (both 27ac and 27me3), Active (27ac alone), Repressed (27me3 alone), and Un-marked (Neither). TSS counts ≥1 were considered positive in expression. Top panel: percentage of CGI-positive (TATA-negative) ex5_cluster per enhancer state; second to bottom panels: CGIap-positive (CGI-positive ex5_clusters that are adjacent to promoter), dCGInap-positive (dCGI-positive ex5_clusters that are not adjacent to promoter) and TATA-positive (CGI-negative). **h**, Enrichment of enhancer states with different classes of ex5_clusters as shown in (**g**). **i**, Same analysis as (**h**), restricted to expressing ex5_clusters per cell-type and incorporating cell-type-specific chromatin states. Unless otherwise stated, statistical comparisons were performed using a two-sided Wilcoxon test. Fisher's exact test was used in (**h**) and (**i**) to calculate odds ratios (OR). $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), n.s. = not significant.